

The Codimension-Two Full Interval for Non-Attaining Projections in $C[0, 1]$

Automatic Math Discovery Smoke Test

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Source and Extracted Question

The source is Tomasz Kania and Grzegorz Lewicki, *Finite-codimensional subspaces of Daugavet spaces: projection constants and minimal projections*, arXiv:2604.10771. Question 6.2 asks whether, for each fixed $n \in \mathbb{N}$, every value in the interval

$$(2, 1 + \lambda_n]$$

occurs as the projection constant of a codimension- n subspace Y of the real space $C[0, 1]$ for which no minimal projection exists. Here λ_n is the maximal absolute projection constant among real n -dimensional Banach spaces.

$$\lambda_n := \sup\{\lambda(E) : \dim E = n\},$$

where $\lambda(E)$ denotes the absolute projection constant of E , that is,

$$\lambda(E) := \sup\{\lambda(E, Z) : E \subset Z \text{ isometrically}\}.$$

The numbers λ_n are classical and notoriously difficult to determine; see [16, 20, 11, 12].

Question 6.2. Fix $n \in \mathbb{N}$. Does every value in the full interval

$$(2, 1 + \lambda_n]$$

occur as the projection constant of a codimension- n subspace Y of the real space $C[0, 1]$ for which no minimal projection exists?

Remark 6.3. If such a space Y existed with

$$\lambda(Y, C[0, 1]) = 1 + \lambda_n,$$

then Theorem 3.1 would force its annihilator

$$W := Y^\perp \subset M[0, 1]$$

to satisfy

$$\lambda(W, M[0, 1]) = \lambda_n.$$

Thus W would have to realise the maximal absolute projection constant λ_n inside $M[0, 1]$.

The present packet settles this question for $n = 2$. We use the classical value $\lambda_2 = 4/3$, attained by the regular hexagon norm.

Proof Intuition

Kania–Lewicki reduce finite-codimensional subspaces of a Daugavet space to finite-dimensional subspaces of the dual. In the codimension-two case it is therefore enough to build two-dimensional

subspaces $W \subset M[0, 1]$ with the desired projection constants and with no w^* -continuous minimal projection.

The useful models are the hyperplanes

$$V_t = \{x \in \ell_1^3 : x_1 + x_2 + tx_3 = 0\}, \quad 0 < t \leq 1.$$

Their projection constants fill the whole interval $(1, 4/3]$. The new point is that no duplication-stability argument is needed. The unique minimal projection onto V_t forces the interval averages of any w^* -continuous minimal projection to land on prescribed exposed faces of the coefficient ball

$$|a + tb| + |a| + |b| \leq r(t).$$

On two adjacent intervals these faces are disjoint. Continuous coefficient functions cannot approach one face from the left and a disjoint face from the right at the common endpoint.

Finite-Dimensional Model

For $0 < t \leq 1$, put

$$D = 1 + t + t^2, \quad r(t) = \frac{(1+t)^2}{D},$$

and define

$$V_t = \ker s_t \subset \ell_1^3, \quad s_t(x_1, x_2, x_3) = x_1 + x_2 + tx_3.$$

Lemma 1. *The space V_t has a unique minimal projection in ℓ_1^3 . It is*

$$P_t = \text{Id} - u_t \otimes s_t, \quad u_t = \left(\alpha, \alpha, \frac{t}{D} \right), \quad \alpha = \frac{1+t}{2D},$$

and

$$\lambda(V_t, \ell_1^3) = \|P_t\| = r(t).$$

Proof. Every projection from ℓ_1^3 onto V_t has the form

$$P_u = \text{Id} - u \otimes s_t, \quad s_t(u) = 1.$$

Let e_1, e_2, e_3 be the coordinate basis and set

$$A_1 = \|e_1 - u\|_1, \quad A_2 = \|e_2 - u\|_1, \quad A_3 = \|e_3 - tu\|_1.$$

Then $\|P_u\| = \max_j A_j$.

Consider the three sign functionals

$$\phi_1 = (1, -1, -1), \quad \phi_2 = (-1, 1, -1), \quad \phi_3 = (-1, -1, 1)$$

and the positive weights

$$\omega_1 = \omega_2 = \frac{t(1+t)}{2D}, \quad \omega_3 = \frac{1}{D}.$$

They sum to 1, and a direct calculation gives

$$\omega_1 \phi_1 + \omega_2 \phi_2 + t\omega_3 \phi_3 = -\frac{t}{D}(1, 1, t).$$

Since $\|\phi_j\|_\infty = 1$, for every admissible u ,

$$\begin{aligned}\|P_u\| &\geq \sum_{j=1}^3 \omega_j A_j \\ &\geq \omega_1 \phi_1(e_1 - u) + \omega_2 \phi_2(e_2 - u) + \omega_3 \phi_3(e_3 - tu) \\ &= 1 + \frac{t}{D} s_t(u) = 1 + \frac{t}{D} = \frac{(1+t)^2}{D} = r(t).\end{aligned}$$

For the stated u_t , one has $s_t(u_t) = 1$, and the three columns of P_t have signs

$$(+, -, -), \quad (-, +, -), \quad (-, -, +),$$

respectively. Hence the above inequalities are equalities, so $\|P_t\| = r(t)$.

It remains to prove uniqueness. If P_u is minimal, equality holds in the weighted estimate. Since all weights are positive and each $A_j \leq r(t)$, we must have $A_j = r(t)$ for each j , and equality must hold in the corresponding sign estimates. In particular,

$$\phi_1(e_1 - u) = r(t), \quad \phi_2(e_2 - u) = r(t).$$

These two equations are

$$1 - u_1 + u_2 + u_3 = r(t), \quad 1 + u_1 - u_2 + u_3 = r(t),$$

so $u_1 = u_2$. The first equation then gives

$$u_3 = r(t) - 1 = \frac{t}{D}.$$

Together with $s_t(u) = 1$, this yields

$$u_1 = u_2 = \frac{1 - tu_3}{2} = \frac{1+t}{2D}.$$

Thus $u = u_t$, and the minimal projection is unique. □

A Copy in $M[0, 1]$

Let $I_1 = (0, 1/3)$, $I_2 = (1/3, 2/3)$, and $I_3 = (2/3, 1)$, and let

$$\mu_j = 3 \mathbf{1}_{I_j} dt \quad (j = 1, 2, 3).$$

The span $E = \text{span}\{\mu_1, \mu_2, \mu_3\} \subset M[0, 1]$ is isometric to ℓ_1^3 . Define

$$W_t = \{z_1 \mu_1 + z_2 \mu_2 + z_3 \mu_3 : z_1 + z_2 + tz_3 = 0\}.$$

Lemma 2. *For every $0 < t \leq 1$,*

$$\lambda(W_t, M[0, 1]) = r(t),$$

but W_t admits no w^ -continuous minimal projection from $M[0, 1]$ onto W_t .*

Proof. The equality of projection constants is immediate from the finite-dimensional model. Restricting any projection $M[0, 1] \rightarrow W_t$ to E gives a projection $E \rightarrow W_t$, so

$$\lambda(W_t, M[0, 1]) \geq \lambda(V_t, \ell_1^3) = r(t).$$

Conversely, the map

$$\nu \mapsto \sum_{j=1}^3 \nu(I_j) \mu_j$$

is a contractive projection from $M[0, 1]$ onto E . Composing it with the minimal projection $E \rightarrow W_t$ gives a projection of norm $r(t)$. Hence the projection constant is exactly $r(t)$.

Suppose, toward a contradiction, that $Q : M[0, 1] \rightarrow W_t$ is a w^* -continuous projection with $\|Q\| = r(t)$. Use the basis

$$v_1 = \mu_1 - \mu_2, \quad v_2 = t\mu_1 - \mu_3$$

of W_t . Since Q has finite rank and is w^* -continuous, there are continuous real functions $g, h \in C[0, 1]$ such that

$$Q\nu = \left(\int g d\nu \right) v_1 + \left(\int h d\nu \right) v_2 \quad (\nu \in M[0, 1]).$$

For $a, b \in \mathbb{R}$, put

$$N_t(a, b) = |a + tb| + |a| + |b|.$$

Then

$$\|av_1 + bv_2\| = N_t(a, b),$$

and applying Q to point masses gives

$$N_t(g(x), h(x)) = \|Q\delta_x\| \leq r(t) \quad (x \in [0, 1]).$$

Let

$$c_j = \left(\int g d\mu_j, \int h d\mu_j \right) \quad (j = 1, 2, 3).$$

The restriction of Q to E is a projection $E \rightarrow W_t$ of norm at most $r(t)$. By Lemma 1, it is the unique minimal projection. Therefore the first two coefficient columns are

$$c_1 = \left(\alpha, \frac{t}{D} \right), \quad c_2 = \left(\alpha - 1, \frac{t}{D} \right), \quad \alpha = \frac{1+t}{2D}.$$

Now define the two linear functionals

$$L_1(a, b) = 2a + (1+t)b, \quad L_2(a, b) = -2a + (1-t)b.$$

For all $a, b \in \mathbb{R}$, $L_1(a, b) \leq N_t(a, b)$ and $L_2(a, b) \leq N_t(a, b)$. Moreover,

$$L_1(c_1) = r(t), \quad L_2(c_2) = r(t).$$

Since c_1 and c_2 are the μ_j -averages of the continuous coefficient curve $x \mapsto (g(x), h(x))$, while $L_i(g(x), h(x)) \leq r(t)$ pointwise, we get

$$L_1(g(x), h(x)) = r(t) \quad \text{for a.e. } x \in I_1,$$

and

$$L_2(g(x), h(x)) = r(t) \quad \text{for a.e. } x \in I_2.$$

By continuity and closedness, at the common endpoint $x = 1/3$ the point

$$(a, b) = (g(1/3), h(1/3))$$

must satisfy

$$N_t(a, b) \leq r(t), \quad L_1(a, b) = r(t), \quad L_2(a, b) = r(t).$$

The last two equalities imply $b = r(t)$. Hence

$$N_t(a, b) = |a + tr(t)| + |a| + r(t) \leq r(t),$$

which forces $a = 0$ and $a = -tr(t)$. This is impossible because $t > 0$ and $r(t) > 0$. Thus no w^* -continuous minimal projection exists. $\square$

Transfer to $C[0, 1]$

Theorem 1. *For every*

$$2 < \Lambda \leq \frac{7}{3},$$

there is a codimension-2 subspace Y of the real space $C[0, 1]$ such that

$$\lambda(Y, C[0, 1]) = \Lambda,$$

and no minimal projection from $C[0, 1]$ onto Y exists.

Proof. The function

$$r(t) = \frac{(1+t)^2}{1+t+t^2}$$

is continuous and strictly increasing on $(0, 1]$, with

$$\lim_{t \downarrow 0} r(t) = 1, \quad r(1) = \frac{4}{3}.$$

Thus for every $\Lambda \in (2, 7/3]$ there exists $t \in (0, 1]$ such that

$$\Lambda = 1 + r(t).$$

Let $W_t \subset M[0, 1] = C[0, 1]^*$ be the two-dimensional space from Lemma 2, and set

$$Y = W_t^\perp \subset C[0, 1].$$

The source paper's Daugavet duality theorem says that for a finite-codimensional subspace $Y = W^\perp$ of a Daugavet space X ,

$$\lambda(Y, X) = 1 + \lambda(W, X^*),$$

and that a minimal projection onto Y exists if and only if there is a w^* -continuous minimal projection from X^* onto W . Applying this with $X = C[0, 1]$ and $W = W_t$, Lemma 2 gives

$$\lambda(Y, C[0, 1]) = 1 + r(t) = \Lambda,$$

and no minimal projection onto Y exists. Since $\dim W_t = 2$, the subspace Y has codimension 2. $\square$

Corollary 1. *Question 6.2 of arXiv:2604.10771 has an affirmative answer for $n = 2$.*

Proof. The classical two-dimensional maximal absolute projection constant is $\lambda_2 = 4/3$. Therefore the interval in Question 6.2 is exactly $(2, 7/3]$, and Theorem 1 realizes every value in it with no minimal projection. $\square$

Novelty Check and Limitations

The cheap run indexes contained only the earlier failed attempt under the slug *codim2 endpoint regular-hexagon route*. That attempt identified the family V_t , but stopped because block duplicates do not have unique minimal projections. The present proof bypasses duplication entirely by using the equality case of the original three-coordinate projection problem and an exposed-face continuity obstruction.

Bounded web searches on 2026-06-29 for the arXiv id, the paper title, codimension-two/full-interval phrases, and the $\ker(1, 1, t)$ projection constant route found no later explicit solution. This novelty check is not a formal literature review.

The theorem settles the fixed-codimension question only for $n = 2$. It does not address the remaining cases $n \geq 3$.

References

- [1] T. Kania and G. Lewicki, *Finite-codimensional subspaces of Daugavet spaces: projection constants and minimal projections*, arXiv:2604.10771, 2026.
- [2] H. König and N. Tomczak-Jaegermann, *Bounds for projection constants and 1-summing norms*, Trans. Amer. Math. Soc. **320** (1990), 799–823.